Quantile Regression Approach in Analysis of Factors Affecting Song Popularity

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***Abstract*—The music industry generates huge revenue from songs. While various studies have been carried out to predict the popularity of the songs, to the best of our knowledge, this is the first study that employs quantile regression, which analyzes various quantiles of data containing songs and the various factors that contribute to their popularity. The analysis may help various songwriters, artists, music industries, and producers determine which types of songs have a higher probability of becoming popular. The high preference of the audience for the song makes a song popular in society. Quantile regression at the 0.10, 0.25, 0.50, 0.75, and 0.90 quantiles is taken to understand the relationship between song popularity and various factors affecting it such as song duration, liveness, acoustics, tempo, danceability, loudness, and others. This may help the music industry do commercial analysis to determine how various factors contribute to making a song popular.**

***Index Terms*—Quantile regression, OLS regression, song popularity, Music Industry**

I. Introduction

Music industries are evolving day by day around the globe with an increase in digital platforms and online mediums. Song preferences by various artificial intelligence-based musical applications provide users with a personalized option to listen. According to the Global music report by the International Federation of the Phonographic Industry for the year 2022, the average music listening time per week is 20.1 hours in 2022 while in the year 2021, it was about 18.4 hours. Music and songs are one of the most engaging activities which people love to spend time on. Introducing diverse songs to the audience through various music streaming markets has bought a new evolution in this field. The music industry has gone through many changes [1]. It is now leading to more advancements and technologies such as paid subscriptions and various artificial intelligence-based personalized applications. Digital distribution of music to various music platforms has increased its reachability to a greater extent. According to the RIAA music revenue report for the first half of 2022, the number of paid subscriptions increased to a record high of 90 million for U.S. recorded music sales, with sales increasing by 10% to $5.0 billion and accounting for more than two-thirds of the first half total. Hence, in the evolving era of music, there is a need to analyze the various factors which are affecting the popularity of the song. Music is also one of the efficient ways to reduce anxiety by providing a great sense of relaxation and also reducing stress [2]. In [3], various trends in the Western music industry were observed, which appeared in loudness, pitch, and timber. Machine learning-based algorithms were used to predict the popularity of the song by comparing six machine learning algorithms(Naïve Bayes, Decision Tree Classifier, Random Forest Classifier, SVM, K-Nearest Neighbors, and Logistic Regression)and used one with the best accuracy. It was observed that the most important factor was tempo, followed by sentiment while the playlist genre and key were discovered to be the least important [4]. Junghyuk Lee et al. [5] investigated what a song's popularity is, how popularity metrics appear in the real world, and also if they are predictable based on acoustic characteristics.

The Quantile Regression analysis in this study will provide various insights to music industries and professionals regarding various strategies, business planning, and financial benefits. Along with that, it will enhance the listener experience.

II. Quantile Regression

The Ordinary Least Squares(OLS) regression is a statistical technique to establish a relationship between predictor variables and the outcome variable [6]. This uses conditional mean to predict the dependent variable. Quantile regression is an extended form of linear regression which uses conditional median to predict the outcome [7]. A quantile defines a particular part of the dataset and the quantile regression examines the relationship between predictor variables and the various quantiles of the outcome variable. It enables us to

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investigate the effect of predictors on different quantiles of the response distribution, providing a comprehensive picture of the distribution.In addition, quantile regression is less susceptible to outliers than the ordinary least squares linear regression model. It can be used when certain conditions of the linear regression model is not met.The sum of weighted absolute residuals is the objective function of this regression, on the basis of various values of τ the different quantiles can be obtained. So at $\tau = 0.50$ the quantile regression is termed as median regression.

The basic ordinary least squares based linear regression model is described below where Y_j represents the jth response, X_{ji} represents regressor level X_i or jth observed value assuming n>p observations are present [8].

$$Y_j = \beta_0 + \beta_1 X_{j1} + \beta_2 X_{j2} + \beta_3 X_{j3} + ... + \beta_p X_{jp} + error_j \quad (1)$$

(j=1,2...n), summarising the equation(1) we get,

$$Y_j = \beta_0 + \sum_{i=1}^{p} \beta_i X_{ji} + error_j$$

where the error term is the difference between the actual and the predicted outcome. β_0, β_1... are the regression coefficients.The squared error sum to be minimized is the following:

$$\sum_{j=1}^{n} (error_j)^2 = \sum_{j=1}^{n} (Y_j - (\beta_0 + \sum_{i=1}^{p} \beta_i X_{ji}))^2 \quad (2)$$

Comparatively, the quantile regression model is an extension of the traditional regression model where a new term τ gets added:

$$Q\tau(Y_j) = \beta_0(\tau) + \beta_1(\tau) X_{j1} + \beta_2(\tau) X_{j2} + \beta_3(\tau) X_{j3} + \beta_4(\tau) X_{j4} ... + \beta_p(\tau) X_{jp} + error_j$$

$Q\tau(Y_j)$: τ –th quantile of dependent variable. So, here we can observe that the value of the coefficients in the above quantile regression equation depends on the τ. The error terms to be minimized in quantile regression are the absolute errors while in the case of ordinary least squares linear regression, it is the sum of squared errors as given in equation (2). Quantile regression seeks to reduce the weighted sum of the negative and positive error terms or the sum of asymmetric penalties that represents the under-prediction and over-prediction described below:

$$\sum_{Y_j < \hat{Y_j}} (1-\tau) |Y_j - \hat{Y_j}| + \sum_{Y_j \geqslant \hat{Y_j}} (\tau) |Y_j - \hat{Y_j}|$$

where

$$\hat{Y_j} = \beta_0(\tau) + \beta_1(\tau) X_{j1} + \beta_2(\tau) X_{j2} + \beta_3(\tau) X_{j3} + \beta_4(\tau) X_{j4} ... + \beta_p(\tau) X_{jp}$$

τ represents the quantile level whose value lies between 0 to 1.

Hence, simply using ordinary least squares linear regression provides limited insights while quantile regression provides the complete picture or relationship at various values of τ. The ordinary least squares linear regression has certain limitations such as it is not robust to outliers, also it confronts homoscedasticity so quantile regression overcomes this. There exist various applications of quantile regression in medicines, science, technology, botany, and many more [9].

III. Methodology

A. Data

The dataset consists of 18,835 songs and their respective song popularity is taken as a response variable to analyze the various factors determining it. Data for this analysis comes from the Kaggle repository, https://www.kaggle.com/datasets/yasserh/song-popularity-dataset, which contains various factors contributing to the popularity of the song. Some of the variables which may be relevant after considering previous studies on the popularity of the songs are considered in our study from the dataset. The explanatory variables taken in our study are acousticness, liveness, loudness, speechiness, tempo, instrumentalness, song duration, danceability, energy, and audio valence.

Song popularity is the dependent variable in the dataset.

TABLE I
INDEPENDENT VARIABLES DEFINITION

Variables	Definition
Xsong_duration_ms	Duration of the song.
Xacousticness	This is indicator of a song's acoustic nature.
Xdanceability	With how ease a person can dance on a song.
Xenergy	A perceptual measure of activity and intensity.
Xinstrumentalness	Measurement of vocals in the song.
Xliveness	Presence of audience while recording.
Xloudness	Method of measuring audio levels based on how humans perceive sound.
Xspeechiness	Detects the no of words spoken in the song.
Xtempo	It is how fast or slow a song is performed.
Xaudio_valence	Describes a track's musical positivity.

Table .I contains the definition of all the independent variables considered in the study. Danceability, loudness, energy, valence, liveness, instrumentalness, acousticness, and speechiness are the float variables whose values lie between 0.0 to 1.0. The duration of the song is an integer variable that is measured in milliseconds.

B. Analysis model

The OLS linear regression model has some assumptions which are required to be met such as there should be linearity in regression coefficients and also in error terms. Fig 1 residual vs fitted plot depicts that the model is not completely linear. The second assumption which fails here is error terms in the model should have constant variance i.e. homoskedasticity. The scale-location plot in Fig 1 shows the error terms do not have constant variance. Also, when Breusch-Pagan Test was applied, the p-value came out to be 2.2e-16, hence, the model has heteroskedasticity as we can reject the null hypothesis. The condition for normal distribution(error terms) should be satisfied, however, it is not a necessary condition as depicted in a normal Q-Q plot. A bit of skewness is depicted in the plot, which shows the distribution is not completely normal. There also exists some influential points near the cook's

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distance in the residual vs leverage plot in Fig 1. However, the reason we have considered quantile regression for the song popularity is not just because of the failure of certain OLS regression assumptions but the quantile regression yields in-depth analysis at various spectrums of quantiles, which provides a comprehensive picture of the predictor and the dependent variables.

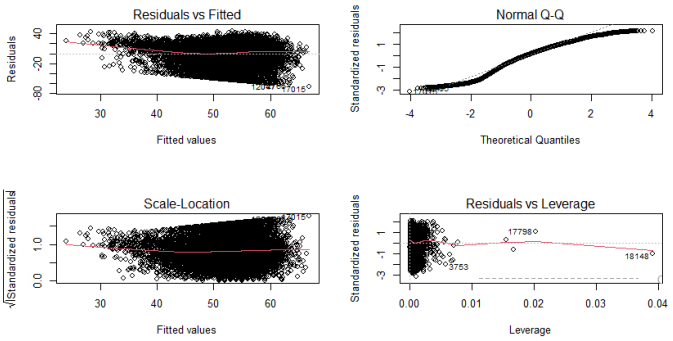


Fig. 1. Residual plots

The model of quantile regression is used to understand the relationship at various quantiles of the dependent variable. It is not possible to analyze the relationship between variables at different distributions if OLS regression is used. Since The OLS-based regression gives a constant coefficient value while the quantile regression gives different values of the coefficients at different quantiles hence, it becomes more appropriate to understand the relationship between response and predictor variables. However, one possibility that comes to mind is to divide the observation based on song popularity and then apply the OLS regression on each model but this results in sample selection biasness [10]. Hence, quantile regression is considered which does not divide the observations randomly. The quantile regression model for song popularity model is represented by the following equation where $\beta_0, \beta_1 ... \beta_9$ represents the regression coefficients and the error term is represented by e.

$$\begin{aligned} songs_popularity = & \beta_0 + \beta_1 X song_duration_ms \\ & + \beta_2 X accousticness + \beta_3 X energy \\ & + \beta_4 X instrumentalness \\ & + \beta_5 X liveness + \beta_6 X loudness \\ & + \beta_7 X speechiness + \beta_8 X tempo \\ & + \beta_9 X audio_valence + e \end{aligned}$$

Here, the variable song popularity is taken as the response variable. The relationship between the response variable and factors that may affect it such as duration of the song, accousticness, and others are considered. For computation, we used RStudio software 2022.12.0 version. To perform the Quantile Regression QuantReg package in R was used.

IV. Results and Discussion

A. Descriptive Statistics

The summary of descriptive analysis on independent and dependent variables is presented in Table .II. The popularity of the song in the dataset lay between 0 to 100 represented by min and max values, the average(mean) is 52.99 in the total of 18,835 songs. The skewness of song duration, instrumentalness, speechiness and liveness was right skewed as the tail was extended to the right side. On average, the loudness of the songs present in the dataset is -7.45 decibels. The duration of the song in the dataset lay between 12000 ms to 1799346 ms whereas the median was 211306 ms. Song duration had a kurtosis value of 46.7 which depicts that it is more peaked(leptokurtic) i.e positive kurtosis. However, other variables also had positive kurtosis such as instrumentalness, liveness, loudness, and speechiness but their values were significantly lesser than the song duration. The mean and median values of the audio valence came out to be equal that is 0.53. The range for danceability and liveness is the same which can be seen from the values of max and min. Table .II shows that on average the danceability factor in songs was more than liveness.

TABLE II
Descriptive Analysis

Variables	Mean	Median	Kurtosis	Skewness	Min	Max	Count
song_popularity	52.99	56.00	-0.17	-0.50	0.00	100.00	18835
song_duration_ms	218211.59	211306.00	46.71	3.26	12000.00	1799346.00	18835
acousticness	0.26	0.13	-0.10	1.07	0.00	1.00	18835
danceability	0.63	0.65	-0.07	-0.39	0.00	0.99	18835
energy	0.64	0.67	-0.14	-0.62	0.00	1.00	18835
instrumentalness	0.08	0.00	7.56	2.99	0.00	1.00	18835
liveness	0.18	0.12	5.79	2.22	0.01	0.99	18835
loudness	-7.45	-6.56	6.52	-1.93	-38.77	1.59	18835
speechiness	0.10	0.06	6.50	2.27	0.00	0.94	18835
tempo	121.07	120.01	-0.22	0.44	0.00	242.32	18835
audio_valence	0.53	0.53	-0.98	-0.02	0.00	0.98	18835

TABLE III
Quantile Regression Model Regression Coefficients.

	10th	25th	50th	75th	90th
Xsong_duration_ms	−0.00001	−0.00001	0.00000	0.00000	−0.00001***
Xacousticness	0.932	−4.773***	−5.604***	−5.837***	−4.741***
Xdanceability	19.824***	11.532***	10.043***	12.622***	18.433***
Xenergy	−10.563***	−12.234***	−13.801***	−11.056***	−8.531***
Xinstrumentalness	−8.232***	−9.127***	−9.745***	−12.947***	−11.233***
Xliveness	−4.649*	−6.149***	−4.610***	−4.628***	−5.328***
Xloudness	0.318***	0.767***	0.804***	0.588***	0.677***
Xspeechiness	−0.509	−5.329**	−5.390***	−0.228	3.973*
Xtempo	−0.014	−0.020**	−0.020***	−0.012**	−0.004
Xaudio_valence	−7.290***	−10.168***	−7.518***	−6.825***	−12.565***
Constant	26.803***	59.821***	74.535***	79.823***	88.868***

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

B. Quantile regression analysis

Table .III shows the quantile regression coefficients after applying the quantile regression at the 10th, 25th, 50th,

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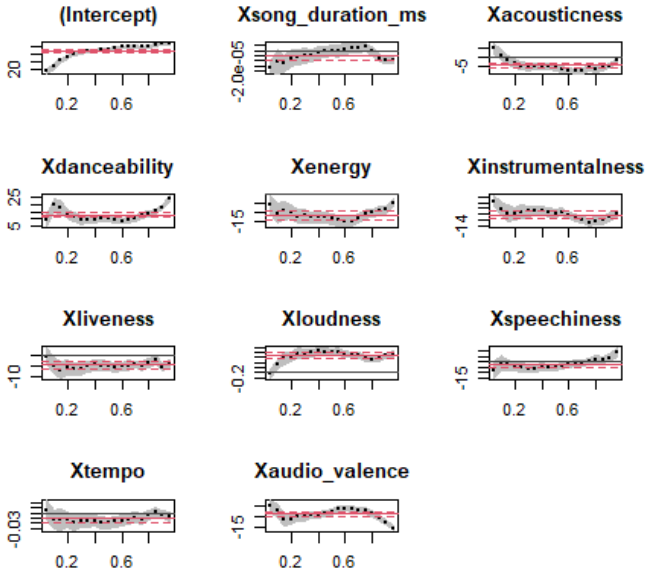


Fig. 2. Coefficient regression graph representing OLS regression and Quantile regression result at 10th, 25th, 50th, 75th, and 90th Quantile.

75th, and 90th quantiles respectively. The result of the OLS regression was also obtained for various independent variables shown in Fig 2. It was observed that the values of the coefficients of the quantile regression for various quantiles were differ from the coefficient of the OLS regression coefficients. The solid red line in Fig2 represents OLS regression and the corresponding dashed line represents the confidence interval at 95%. We can observe that coefficients in OLS regression remain constant at all the quantiles so there exists a straight line. While the black dotted line represents the Quantile regression coefficients and their respective confidence interval in the shaded region. The danceability and loudness were significant and positive at all levels in the Quantile regression. The impact of danceability (50th: 10.043, 75th:12.622, 90th:18.433)indicated that after the 50th quantile, the danceability coefficient was increasing i.e at upper quantiles danceability factor impacted more than the middle quantile, this states that danceability impacts more to more popular songs. Liuchang xu et al. [11] concluded that most people do not enjoy sad music. Also according to paper [12], danceable songs are preferred. It is difficult to analyze the popularity of the song at various distributions based on the duration of the song because at all the quantile levels the coefficient is almost the same but it is significant at the higher quantile. The loudness of the song was maximum at the 50th quantile, hence if a song was not most popular and also not least popular i.e average popular songs had high loudness. According to the study in [13], loudness is one of the major factors behind making a song hit. Chinese Listeners as per [14] preferred more rock and pop kind of music which is usually louder. This study to the best of our knowledge is the first application of quantile regression to be applied to analyze

TABLE IV
SLOPE EQUALITY TEST

Test Summary	Chi. Sq. Sta	Chi-Sq. d.f.	Prob.
Wald Test	124.29	40	0.0000

the determinants of the popularity of a song. Z Al-Beitawi et al. [15] used cluster analysis of musical attributes to decide which song will trend also resulting that danceability being one of the significant factors. Acousticness was significant at all the quantiles and had a negative correlation while at the 10th quantile, it was insignificant. Thus, we may say that the songs which were not popular had no effect on acoustic nature. Liveness was significant but negatively associated with the song's popularity in all quantiles, which states that to create a popular song it should not be recorded live as live recorded music may contain noise. At the 90th quantile of the dependent variable, the coefficient of regression was -5.328 for liveness, which states that a unit increase in liveness reduces song popularity by 5.328 keeping other explanatory variables constant. Audio valence and instrumentalness had a significantly negative effect on the popularity of the song. The negative region effect was higher at more popular songs than the less popular songs for instrumentalness. Green Arther Sandag and Adinda Marcellina Manueke in [16] used the decision tree algorithm for group and solo singers dataset in the predictive model and also concluded that the instrumental songs were less attractive while energy and specific loudness may contribute to it.

As per Chi-Square statistics done in Wald test the value of the slope equality test is 124.29, Slope equality hypothesis is rejected at a 5% significance level, which means that the slope equality is not the same at all the quantiles and it is statistically significant as shown in Table .IV.

According to our quantile regression analysis, songs with higher popularity had less audio valence as compared to the songs with lower popularity as represented in Fig 2. As we are moving from the lower quantile to the higher quantile, the effect of audio valence becomes lesser which depicts that an extremely euphoric track is less attractive to music listeners, there should be a good balance. Audio valence is not influential in determining the popularity of songs as given in the paper [17]. However, as per work done by Yutong Ge et al. [18] valence and speechiness were correlated with the popularity of the song. Energy in a song had however inverse results as compared to audio valence. Energy also had a negative sign on the popularity of the song. As presented in paper [19], the energy is positively related to the popularity of the song. Music makers should consider that more energetic songs had a higher negative effect on more popular songs. Research done in [20] used the CART decision tree to study the Indonesian music market and also resulted that medium energy and moderate valence makes a song more popular. The coefficient of OLS regression can be seen and compared with coefficients at various quantiles in Fig 2. We can see

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that it changes at every quantile hence, gives us a way to do analysis at various distributions. According to coefficients of quantile regression, the loudness of a song had a significant and positive effect on all the quantiles. The effect of loudness was maximum at the 50th quantile i.e average popular songs was louder which may be because loud music is not enjoyed by people of different age groups and as per result the less popular songs had less loudness. The Speechiness of the song is significantly positive at the 90th quantile which shows that the songs which are more popular contain more spoken words in the song as compared to the music since, in a popular song if spoken words are more, it may keep the listener busy and hence the popularity may increase. The tempo of the song was negatively related to its popularity of the song. The tempo was significant at the 20th, 50th, and 75th quantiles which state that tempo effect on song popularity may vary across the quantiles. L Colley et al. [21] took various digital platforms to study the characteristics involved in the popularity of the song and in one of them they also found that instrumentalness and acousticness are negatively related to the popularity of the song. In [22] researchers used the Million Song Dataset which contains about one million songs' audio features, to study the important factors in the popularity of songs using regression algorithms and classification techniques. The Goodness of Fit test (suggested in [23]) used in this Quantile regression model resulted that for the 90th quantile, it fits the best with the pseudo-R-square value of 0.07. The pseudo r square test for 10th,25th,50th, and 75th quantiles are 0.011, 0.016, 0.024, and 0.040 respectively. It also states that the 90th quantile better fits the model as compared to other quantiles.

V. Conclusion and Future Work

This study used quantile regression analysis to identify the determinants of the popularity of the songs. The traditional OLS regression does not provide a detailed analysis of various distributions of dependent variables. Data from about 18,835 songs were taken. In particular, the study employed quantile regression analysis that showed the effects of variables such as tempo, audio valence, danceability, liveness, loudness, energy, acoustics, instrumentals, song duration, and speechiness in each song popularity quantile to identify the effect of each variable. Quantile regression is helpful for the detailed analysis of various quantiles of song popularity. According to our analysis, danceability and loudness were significant and had a positive effect on all the quantiles. The danceability factor was high for more popular songs which states that highly popular songs were more danceable. For average popular songs(50th quantile)loudness effect was greater i.e average popular songs were louder as compared to the least and most popular songs. The duration of the song had negative relation and didn't varies across the quantiles, hence it is not possible to distinguish the effect at various quantiles. This analysis may help the music industry and song makers to generate maximum revenue and consider the involved factors to make their songs popular, for the least popular songs acoustics and tempo did not matter as they were insignificant. The effect of energy was more at higher quantiles also it was negatively related to the song's popularity.

In future work, categories of different song genres and languages can be studied separately for further analysis and deriving patterns to understand the variables in detail.

References

[1] J. R. Ogden, D. T. Ogden, and K. Long, "Music marketing: A history and landscape," *Journal of Retailing and Consumer Services*, vol. 18, no. 2, pp. 120–125, 2011.

[2] E. Labbé, N. Schmidt, J. Babin, and M. Pharr, "Coping with stress: the effectiveness of different types of music," *Applied psychophysiology and biofeedback*, vol. 32, pp. 163–168, 2007.

[3] J. Serrà, Á. Corral, M. Boguñá, M. Haro, and J. L. Arcos, "Measuring the evolution of contemporary western popular music," *Scientific reports*, vol. 2, no. 1, pp. 1–6, 2012.

[4] J. Kamal, P. Priya, M. Anala, and G. Smitha, "A classification based approach to the prediction of song popularity," in *2021 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES)*, pp. 1–5, IEEE, 2021.

[5] J. Lee and J.-S. Lee, "Music popularity: Metrics, characteristics, and audio-based prediction," *IEEE Transactions on Multimedia*, vol. 20, no. 11, pp. 3173–3182, 2018.

[6] V. Lechene, K. Pendakur, and A. Wolf, "Ordinary least squares estimation of the intrahousehold distribution of expenditure," *Journal of Political Economy*, vol. 130, no. 3, pp. 681–731, 2022.

[7] R. Koenker and G. Bassett Jr, "Regression quantiles," *Econometrica: journal of the Econometric Society*, pp. 33–50, 1978.

[8] D. C. Montgomery, E. A. Peck, and G. G. Vining, *Introduction to linear regression analysis*. John Wiley & Sons, 2021.

[9] J. Chen and J. Ding, "A review of technologies on quantile regression," in *Statistics and Information Forum*, vol. 23, pp. 89–96, 2008.

[10] J. Heckman *et al.*, "Sample selection bias as a specification error," *Applied Econometrics*, vol. 31, no. 3, pp. 129–137, 2013.

[11] L. Xu, Y. Zheng, D. Xu, and L. Xu, "Predicting the preference for sad music: the role of gender, personality, and audio features," *IEEE Access*, vol. 9, pp. 92952–92963, 2021.

[12] M. Interiano, K. Kazemi, L. Wang, J. Yang, Z. Yu, and N. L. Komarova, "Musical trends and predictability of success in contemporary songs in and out of the top charts," *Royal Society open science*, vol. 5, no. 5, p. 171274, 2018.

[13] Y. Ni, R. Santos-Rodriguez, M. Mcvicar, and T. De Bie, "Hit song science once again a science," in *4th International Workshop on Machine Learning and Music*, 2011.

[14] Q. Shen, H. Zhang, and J. Cai, "Chinese listeners' preferences of american and european pop music and comparison with the world trends," in *2019 IEEE/ACIS 18th International Conference on Computer and Information Science (ICIS)*, pp. 325–328, IEEE, 2019.

[15] Z. Al-Beitawi, M. Salehan, and S. Zhang, "What makes a song trend? cluster analysis of musical attributes for spotify top trending songs," *Journal of Marketing Development and Competitiveness*, vol. 14, no. 3, pp. 79–91, 2020.

[16] G. A. Sandag and A. M. Manueke, "Predictive models for popularity of solo and group singers in spotify using decision tree," in *2020 2nd International Conference on Cybernetics and Intelligent System (ICORIS)*, pp. 1–5, IEEE, 2020.

[17] Y. Yee and M. Raheem, "Predicting music popularity using spotify and youtube features," *Indian Journal of Science and Technology*, vol. 15, no. 36, pp. 1786–1799, 2022.

[18] Y. Ge, J. Wu, and Y. Sun, "Popularity prediction of music based on factor extraction and model blending," in *2020 2nd International Conference on Economic Management and Model Engineering (ICEMME)*, pp. 1062–1065, IEEE, 2020.

[19] D. P. D. Singh, "How to make a hit song: Invest in a good studio," *Available at SSRN 3939762*, 2021.

[20] L. R. Febirautami, I. Surjandari, and E. Laoh, "Determining characteristics of popular local songs in indonesia's music market," in *2018 5th International Conference on Information Science and Control Engineering (ICISCE)*, pp. 197–201, IEEE, 2018.

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[21] L. Colley, A. Dybka, A. Gauthier, J. Laboissonniere, A. Mougeot, N. Mowla, K. Dick, H. Khalil, and G. Wainer, "Elucidation of the relationship between a song's spotify descriptive metrics and its popularity on various platforms," in *2022 IEEE 46th Annual Computers, Software, and Applications Conference (COMPSAC)*, pp. 241–249, IEEE, 2022.

[22] J. Pham, E. Kyauk, and E. Park, "Predicting song popularity," *Dept. Comput. Sci., Stanford Univ., Stanford, CA, USA, Tech. Rep*, vol. 26, 2016.

[23] R. Koenker and J. A. Machado, "Goodness of fit and related inference processes for quantile regression," *Journal of the american statistical association*, vol. 94, no. 448, pp. 1296–1310, 1999.

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